

Week 6 — Deterministic Location–Allocation Problem (LAP)

OL7014 Integer Programming • Instructor: Prof. H. Park • Week of Sep 28, 2026

1. Facility location — paying to open

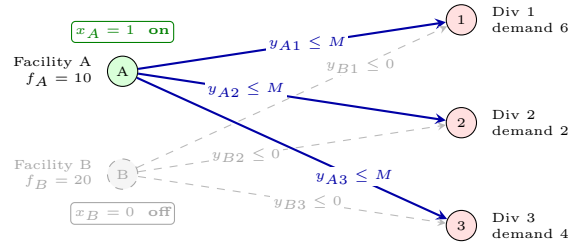
week 5's transportation assumed the facilities existed — *which facilities should exist?*

watch the letters: back to week 2's convention — x = integer (open or not), y = continuous. relief tons are divisible, so the flow is no longer the integer decision

$x_i = 1$ iff facility i opens, f_i = fixed opening cost, y_{ij} = tons shipped

$$\begin{aligned} \min_{x,y} \quad & \sum_i f_i x_i + \sum_i \sum_j c_{ij} y_{ij} \quad \text{fixed + shipping} \\ \text{s.t.} \quad & \sum_i y_{ij} = d_j \quad (\text{demand}) \\ & y_{ij} \leq M x_i \quad (\text{linking — week 2}) \\ & y_{ij} \geq 0, \quad x_i \in \{0, 1\} \end{aligned}$$

$x_i = 0 \Rightarrow y_{ij} = 0$; $x_i = 1$ releases it — *uncapacitated*



week 5 §3's graph, nothing added but a **switch** $x_i \in \{0, 1\}$ on each facility, price f_i
B off $\Rightarrow y_{Bj} \leq M \cdot 0$: its three roads carry nothing — the linking row is the switch

how big is M ? $y_{ij} \leq d_j$ always $\Rightarrow M = d_j$, smallest valid. $M = \sum_j d_j$: also valid, but looser — the LP bound sags with it (week 2: the *smallest* valid M)

1. **fix $x \Rightarrow$ transportation falls out: all the hardness is in the binaries**

2. **big- M costs the guarantee:** $y_{ij} \leq M x_i$ belongs to no incidence matrix $\Rightarrow$ week 5's MCNF proposition is gone. (not TU $\neq$ the LP lies: only *some* b must go fractional. §2 checks)

KEY every guarantee in week 5's note is a property of A — so every one is a single modeling decision away from being lost.

2. Deterministic LAP — MCNF + Facility Location

Remember week 5's conservation row (MCNF)? the flow is now called y — nothing else changes:

$$\sum_j y_{ij} - \sum_j y_{ji} = b_i \quad (\forall i \in V) \quad \text{outflow} - \text{inflow} = \text{net supply}$$

what is b_i at a node? stock placed there, minus demand — and two slacks so the row can always balance:

$$b_i = \underbrace{r_i}_{\text{decision}} - \underbrace{d_i}_{\text{data}} + q_i - s_i$$

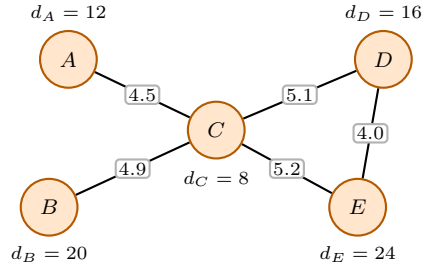
$r_i \geq 0$ = reserve held at i (cost h_i /ton), $q_i \geq 0$ = unmet demand (penalty u_i), $s_i \geq 0$ = surplus (penalty o_i ; not transportation's supply)

two scalars of *data*: B = the stock budget (tons available to preposition), p = the most facilities we may open

$$\begin{aligned} \min_{x,r,y,q,s} \quad & \sum_{i \in V} f_i x_i + \sum_{i \in V} h_i r_i + \sum_{(i,j) \in E} c_{ij} y_{ij} + \sum_{i \in V} u_i q_i + \sum_{i \in V} o_i s_i \\ \text{s.t.} \quad & \sum_j y_{ij} - \sum_j y_{ji} = r_i - d_i + q_i - s_i & (\forall i \in V) & \text{[MCNF's conservation]} \\ & \sum_{i \in V} r_i \leq B & & \text{[budget: a knapsack row]} \\ & r_i \leq B x_i & (\forall i \in V) & \text{[linking — week 2]} \\ & \sum_{i \in V} x_i \leq p & & \text{[at most } p \text{ facilities]} \\ & y, r, s \geq 0, \quad 0 \leq q_i \leq d_i, \quad x_i \in \{0, 1\} \end{aligned}$$

five costs: open (f_i), hold (h_i), ship (c_{ij}), unmet demand penalty (u_i), surplus maintenance cost (o_i). every constraint is old

Numerical example



data — everything the model below needs:
 $V = \{A, B, C, D, E\}$
 $E = \{(A, C), (C, A), (B, C), (C, B), (C, D), (D, C), (C, E), (E, C), (D, E), (E, D)\}$
 $d_A = 12, d_B = 20, d_C = 8, d_D = 16, d_E = 24$
 $c_{AC} = 4.5, c_{BC} = 4.9, c_{CD} = 5.1,$
 $c_{CE} = 5.2, c_{DE} = 4.0; c_{ij} = c_{ji}$ on every road
 $f_i = 100, h_i = 10, u_i = 60, o_i = 1$ (all i)
 $p = 2, B = 55$

that map, priced:

$$\min \underbrace{100 \sum_i x_i}_{\sum_{i \in V} f_i x_i} + \underbrace{10 \sum_i r_i}_{\sum_{i \in V} h_i r_i} + \underbrace{60 \sum_i q_i}_{\sum_{i \in V} u_i q_i} + \underbrace{\sum_i s_i}_{\sum_{i \in V} o_i s_i} + \underbrace{4.5(y_{AC} + y_{CA}) + 4.9(y_{BC} + y_{CB}) + 5.1(y_{CD} + y_{DC}) + 5.2(y_{CE} + y_{EC}) + 4.0(y_{DE} + y_{ED})}_{\sum_{(i,j) \in E} c_{ij} y_{ij}}$$

$$\begin{aligned} \text{s.t. } & y_{AC} - y_{CA} = r_A - 12 + q_A - s_A && [\text{flow conservation (node A)}] \\ & y_{BC} - y_{CB} = r_B - 20 + q_B - s_B && [\text{flow conservation (node B)}] \\ & (y_{CA} + y_{CB} + y_{CD} + y_{CE}) - (y_{AC} + y_{BC} + y_{DC} + y_{EC}) = r_C - 8 + q_C - s_C && [\text{flow conservation (node C)}] \\ & (y_{DC} + y_{DE}) - (y_{CD} + y_{ED}) = r_D - 16 + q_D - s_D && [\text{flow conservation (node D)}] \\ & (y_{EC} + y_{ED}) - (y_{CE} + y_{DE}) = r_E - 24 + q_E - s_E && [\text{flow conservation (node E)}] \\ & \sum_i r_i \leq 55 && [\text{budget}] \\ & r_A \leq 55 x_A && [\text{linking (node A)}] \\ & r_B \leq 55 x_B && [\text{linking (node B)}] \\ & r_C \leq 55 x_C && [\text{linking (node C)}] \\ & r_D \leq 55 x_D && [\text{linking (node D)}] \\ & r_E \leq 55 x_E && [\text{linking (node E)}] \\ & \sum_i x_i \leq 2 && [\text{at most } p = 2 \text{ facilities}] \\ & y, r, s \geq 0, \quad 0 \leq q_i \leq d_i, \quad x_i \in \{0, 1\} \end{aligned}$$

demo Deterministic-LAP-Demo.exe